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Frau Lisa Merkel
Pathologisches Institut — Zentrum für Personalisierte Medizin (ZPM)
Universitätsklinikum Heidelberg
Im Neuenheimer Feld 224, 69120 Heidelberg
Job-ID: V000015540

Re: Data Manager*in — Onkologie, ZPM Heidelberg (V000015540)

Dear Frau Merkel,

I am applying for the Data Manager position (V000015540) at the ZPM Heidelberg. My interest in this role is specific: the ZPM operates at the interface between clinical oncology and translational research, and that is precisely the context in which the data management problems are hardest. In purely clinical settings, data flows are standardised. In purely research settings, data structures are investigator-defined. At the interface, neither set of conventions applies cleanly, and managing data quality requires understanding what the data will be used for — not just that it has been entered correctly.

Oncology data infrastructure at clinical scale. Over the past four years I have built the analytical layer that sits downstream of clinical data collection in oncology. The Syndrome framework processes patient and tumour characterisation data to produce neoantigen predictions, MHC binding assessments, immune escape scores, and a structured disease state representation $\mathbf{D} \in [0, 1]^8$ with dimensions including immune activation, mucosal barrier integrity, metabolic, and circadian components. The Gospel pipeline handles genomic variant calling, somatic mutation analysis, RNA-seq quantification, and multi-omics integration from clinical cohort data, validated against gnomAD, 1000 Genomes, and UK Biobank at 10^6 variants per second. The Lavoisier pipeline processes mass spectrometry data (> 100 GB mzML) from clinical samples with 98.9 % NIST classification accuracy. Building these systems required understanding, at each step, how clinical data is generated, what the quality failure modes are, and what the downstream analysis requires from the upstream documentation. That is the same understanding the ZPM data manager role requires in the other direction: knowing what the research and certification use cases need in order to maintain the right data quality upstream.

Clinical data experience. At Universitätsklinikum Hamburg-Eppendorf (2017–2018) I worked directly in the Glycomics and Proteomics laboratory, processing clinical samples, operating automated LC-MS/MS and MALDI-TOF pipelines, and managing data annotation and quality control for clinical cohort studies. At TUM / Bayerisches Forschungsinstitut (2019–2021) I worked on multi-omics data integration across federated clinical sites, maintaining distributed data processing pipelines and managing data quality across heterogeneous institutional data structures. I am familiar with the practical constraints

of clinical data: incomplete records, inconsistent coding, retrospective vs. prospective collection differences, and the documentation requirements for certification and reporting that differ from what researchers want.

eCRF development and EDC systems. I do not have prior experience with Onkostar specifically, and I want to be direct about that. What I bring instead is deep familiarity with the data structures that tumour documentation systems capture and the analysis pipelines that consume them. eCRF design is a data modelling problem: the form must capture exactly what the downstream analysis requires, in a structure that clinicians can complete reliably under time pressure. I have made those modelling decisions from the analysis side many times, and I expect the learning curve on Onkostar's specific interface to be short.

Scientific publications and reporting. The position includes contributing to scientific publications and certification reporting. I have written formal manuscripts with full mathematical apparatus — theorem statements, validation experiments, quantitative performance claims — and I am comfortable with the standards of precision clinical research publications require. For the ZPM context, this means I can contribute to translational manuscripts that span clinical data, genomics, and computational analysis, not just to methods sections.

Interdisciplinary coordination. The ZPM brings together pathologists, oncologists, computational researchers, and IT infrastructure teams. The data manager sits at the centre of that coordination. My work has required continuous communication across exactly this range of expertise: explaining computational methods to clinicians, explaining clinical constraints to software engineers, and translating between the two in project planning and documentation. User training and support — another component of this role — are familiar from deploying tools that wet-lab and clinical collaborators use without local computational infrastructure.

Background. I hold an MSc in Computational Biology (Universität Tübingen), an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and a BSc in Biochemistry and Cell Biology (Jacobs University Bremen). I conducted doctoral research in Computational Lipidomics at TUM (2019–2021). I have lived in Germany continuously since 2011. English is native; German is conversational. I am available immediately.

Yours sincerely,

Kundai Farai Sachikonye